1.3 Chemical Bonding

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.3 Chemical Bonding
Difficulty	Medium

Time allowed: 80

Score: /59

Percentage: /100

Question 1a

Ammonia, NH_3, and methane, CH_4, are the hydrides of elements which are next to one another in the Periodic Table.

In the boxes below, draw the 'dot-and-cross' diagram of a molecule of each of these compounds.

Show outer electrons only.

State the shape of each molecule.

NH_3	CH_4
shape	shape

[3 marks]

Question 1b

Ammonia is polar whereas methane is non-polar. The physical properties of the two compounds are different.

i)
Explain, using ammonia as the example, the meaning of the term bond polarity.

[2]

ii)
Explain why the ammonia molecule is polar.

[1]

[3 marks]

Question 1c

State **one** physical property of ammonia which is caused by its polarity.

[1 mark]

Question 1d

When ammonia gas is mixed with hydrogen chloride, white, solid ammonium chloride is formed.

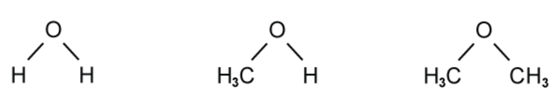
State each type of bond that is present in one formula unit of ammonium chloride and how many of each type are present.

You may draw diagrams.

[3 marks]

Question 2a

The structural formulae of water, methanol and methoxymethane, CH_3OCH_3, are given below.



i)
How many lone pairs of electrons are there around the oxygen atom in methoxymethane?

[1]

ii)
Suggest the size of the C–O–C bond angle in methoxymethane.

[1]

[2 marks]

Question 2b

The physical properties of a covalent compound, such as its melting point, boiling point, vapour pressure, or solubility, are related to the strength of attractive forces between the molecules of that compound.

These relatively weak attractive forces are called intermolecular forces. They differ in their strength and include the following.

A	interactions involving permanent dipoles
B	interactions involving instantaneous dipole-induced dipoles
C	hydrogen bonds

By using the letters A, B, or C, state the strongest intermolecular force present in each of the following compounds.

For each compound, write the answer on the dotted line:

ethanal	CH_3CHO	....
ethanol	CH_3CH_2OH	....
methoxymethane	CH_3OCH_3	....
2–methylpropane	$(CH_3)_2CHCH_3$	....

[4 marks]

Question 2c

Methanol and water are completely soluble in each other.

i)

Which intermolecular force exists between methanol molecules and water molecules that makes these two liquids soluble in each other?

[1]

ii)

Draw a diagram that clearly shows this intermolecular force. Your diagram should show any lone pairs or dipoles present on either molecule that you consider to be important.

[3]

[4 marks]

Question 2d

When equal volumes of ethoxyethane, $C_2H_5OC_2H_5$, and water are mixed, shaken, and then allowed to stand, two layers are formed.

Suggest why ethoxyethane does not fully dissolve in water. Explain your answer.

[2 marks]

Question 3a

This question is about electronegativity.

The electronegativities of some elements are shown in Table 3.1 below.

Table 3.1

Element	Electronegativity
Li	1.0
H	2.1
C	2.5
N	3.0
Cl	3.0

Define the term electronegativity.

[2 marks]

Question 3b

Use Table 3.1 to explain the trend in electronegativity across the Periodic Table.

[3 marks]

Question 3c

Explain how the carbon-hydrogen bond (such as in CH_4) differs from the nitrogen-hydrogen bond (such as in NH_3) in terms of the bond polarity.

[2 marks]

Question 3d

Explain, in terms of electronegativity, why the bonding in ammonia (NH_3) is covalent but the bonding in lithium chloride (LiCl) is ionic.

[4 marks]

Question 4a

Both lithium and lithium chloride contain ions of lithium. However, the structure bonding and properties of these substances are very different.

State how the ions are held together in solid lithium and in solid lithium chloride.

[2 marks]

Question 4b

Table 4.1 below shows the melting and boiling points of lithium and lithium chloride.

Table 4.1

	Melting point (°C)	Boiling point (°C)
Lithium	180.5	1342
Lithium chloride	605.0	1382

Explain what can be deduced from the information in Table 4.1.

[2 marks]

Question 4c

Two students, **A** and **B**, are comparing the properties of lithium and lithium chloride.

Student **A** states that both lithium and lithium chloride will conduct electricity, but Student **B** states that **only** lithium will conduct electricity.

State whether Student **A**, Student **B**, or neither student is correct. Explain your answer.

[5 marks]

Question 4d

Student **A** and **B** then went on to discuss the bonding in another substance, CaF_2.

Explain, in terms of electrons, how CaF_2 is formed from its atoms.

[3 marks]

Question 5a

Alkenes contain a carbon to carbon double bond that consists of a σ bond and a π bond.

Complete the diagram to show the areas of electron density for each bond.
Label the σ bond and the π bond.

[2 marks]

Question 5b

Use the information in Table 5.1 to explain why ethene contains stronger covalent bonds between carbons atoms than ethane.

Table 5.1.

Bond	Bond Length
C-C	150 pm
C=C	134 pm

[3 marks]

Question 5c

Bond length and bond energies can be used to compare the reactivity of covalent molecules.

Compare the reactivity of hydrogen halides using the information in Table 5.2.

Table 5.2

	Bond energy	Bond length (pm)
HCl	431	127
HBr	366	141
HI	299	161

[3 marks]

Question 5d

The boiling points of the hydrogen halides are shown in Table 5.3.

Table 5.3

Hydrogen halide	Boiling Point (K)
H-F	293
H-Cl	188
H-Br	207
H-I	238

i)
Explain why hydrogen halides have relatively low boiling points despite having strong covalent bonds

[2]

ii)
Explain why HF has a higher boiling point than HCl.
You should refer to the van der Waals' forces found in each substance in your answer.

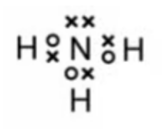
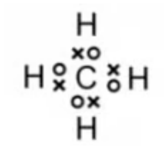
[4]

[6 marks]

Model Answers: Medium

1a

a) The 'dot-and-cross' diagram of a molecule of each of these compounds and their shape is:

NH_3	CH_4
	
pyramidal	tetrahedral

- Both dot and cross diagrams correct; [1 mark]
- NH_3 is pyramidal
  OR
  Trigonal pyramidal; [1 mark]
- CH_4 is tetrahedral; [1 mark]

[Total: 3 marks]

- **Remember:** Both compounds form covalent bonds as they share pairs of electrons
- After drawing your dot and cross diagram, count around each atom to ensure it has a full outer shell of electrons (eight for nitrogen and carbon, two for hydrogen)
- Following VESPR rules:
  - Lone pair-bond pairs will repel each other more than bond pair-bond pairs
    - Ammonia has one lone pair and three bonding pairs so adopts a pyramidal shape with bond angles between the N and H atoms of 107°
    - Methane has four bonding pairs which repel each other equally to give a tetrahedral shape and bond angles between the N and H atoms of 109.5°

1b

b)

i) The meaning of the term bond polarity is:

- Nitrogen and hydrogen have different electronegativities
  OR
  N-H bond has a dipole; [1 mark]
- $N^{\delta-}$ - $H^{\delta+}$
  OR
  Bonding is unequally shared; [1 mark]

ii) The ammonia molecule is polar because:

- The molecule is not symmetrical
  OR
  Dipoles do not cancel out; [1 mark]

[Total 3 marks]

- **Careful:** You must reference ammonia in your answer or you will not score the marks
- Electronegativity is the ability of an atom to attract electrons to itself in a covalent bond
  - The more electronegative the element, the greater its ability to do this
- Electronegativity increases going across and up the Periodic Table
  - Nitrogen has an electronegativity value of 3.0
  - Hydrogen has an electronegativity value of 2.1
  - Nitrogen will attract the electrons in the covalent bond to itself more strongly resulting in unequal electron distribution in the bond resulting in it being polar overall
  - NH_3 is not a symmetrical molecule, so the dipole moments of each N-H bond cannot cancel each other out

1c

c) **One** physical property of ammonia which is caused by its polarity is:

- Ammonia has a higher boiling point than expected;

  OR

It has a higher boiling point than methane;

OR

It is soluble in water; [1 mark]

[Total 1 mark]

- Ammonia, having a simple molecular structure will have instantaneous dipole-induced dipole forces
- Due to being polar, it will also have permanent dipole-permanent-dipole forces, the attractive force between δ+ on one molecule and δ– on another molecule
    - These forces require more energy to overcome than instantaneous dipole-induced dipole forces
- Methane will only have instantaneous dipole-induced dipole forces as the dipole moments created by the difference in electronegativities of the carbon and hydrogen atoms cancel each other out due to the symmetrical nature of the molecule
- To dissolve in water, the lone pair of electrons on the oxygen atoms in a water molecule will be attracted to δ+ hydrogen atom in the ammonia molecule

1d

d) The type of bond that is present in one formula unit of ammonium chloride and number are:

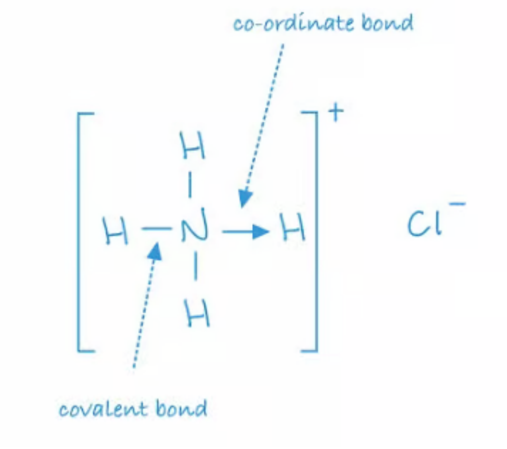
- Three covalent N-H bonds; [1 mark]
- One co-ordinate (dative covalent) N-H bond; [1 mark]
- One ionic bond between NH_4^+ and Cl^-; [1 mark]

[Total: 3 marks]

- There are already three covalent bonds present in NH_3 as seen in part a)
- A co-ordinate bond is one in which both electrons in the covalent bond are provided by one atom
    - Nitrogen has one lone pair of electrons so forms a co-ordinate bond with the H^+ ion from the HCl
    - A co-ordinate bond is represented using an arrow, starting from the atom with the lone pair
    - The resulting ammonium ion has a positive charge as only the nucleus of the

hydrogen was transferred, the electrons from hydrogen remain with the chloride ion

- The negative chloride ion then forms an ionic bond with the ammonium ion



2a

a)

i) The number of lone pairs of electrons around the oxygen atom in methoxymethane is:

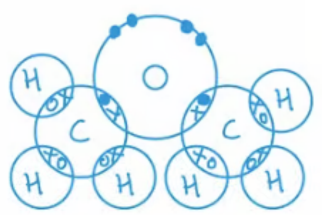
- 2; [1 mark]

ii) The size of the C–O–C bond angle in methoxymethane is:

- Between 104° and 105°; [1 mark]

[Total: 2 marks]

- The arrangement of electrons in methoxymethane is:



- **Careful:** The question asks for the number of **pairs**, not the number in total
- There are two lone pairs and two bonding pairs of electrons
    - The lone pairs will repel each other more than the bonding pairs according to

VSEPR rules resulting in the molecule having a bent linear shape and C-O-C bond angles of 104.5°

2b

b) The strongest intermolecular force in each compound is:

- ethanal CH_3CHO = **A**; [1 mark]
- ethanol CH_3CH_2OH = **C**; [1 mark]
- methoxymethane CH_3OCH_3 = **A**; [1 mark]
- 2–methylpropane $(CH_3)_2CHCH_3$ = **B**; [1 mark]

[Total: 4 marks]

- The order of strength of intermolecular forces beginning with the strongest is:
  - Hydrogen bonding > permanent dipole- permanent dipole > temporary or induced dipoles
- All molecules have temporary or induced dipoles
- Permanent dipole- permanent dipole interactions will arise in ethanal and methoxymethane, due to both molecules being polar
  - The C=O bond in ethanal carries a δ- on the oxygen and δ+ on the carbon
  - The C-O bond in methoxymethane again carried a δ- on the oxygen and δ+ on the carbon

b) The strongest intermolecular force in each compound is:

- ethanal CH_3CHO = **A**; [1 mark]
- ethanol CH_3CH_2OH = **C**; [1 mark]
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[Total: 4 marks]

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  - Hydrogen bonding > permanent dipole- permanent dipole > temporary or induced dipoles
- All molecules have temporary or induced dipoles
- Permanent dipole- permanent dipole interactions will arise in ethanal and methoxymethane, due to both molecules being polar
  - The C=O bond in ethanal carries a δ- on the oxygen and δ+ on the carbon
  - The C-O bond in methoxymethane again carried a δ- on the oxygen and δ+ on the carbon
- Hydrogen bonding will occur in ethanol as it has a lone pair of electrons on the oxygen atom which will form a hydrogen bond with the δ+ hydrogen atom of a neighbouring ethanol molecule

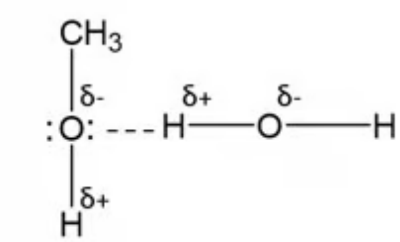
2c

c)

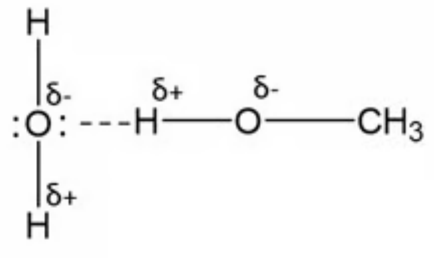
i) The intermolecular force which exists between methanol molecules and water molecules that makes these two liquids soluble in each other is:

- Hydrogen bonding / bonds; [1 mark]

ii) A diagram that clearly shows this intermolecular force is:



OR



- Correct dipoles on an O-H; [1 mark]
- Lone pair on O atom of CH_3OH
  OR
  H_2O clearly shown in the hydrogen bond; [1 mark]
- Hydrogen bond shown between the lone pair of an O atom and a H atom in an -OH bond; [1 mark]

[Total: 4 marks]

- Showing hydrogen bonding between different molecules comes up frequently so invest the time in learning these diagrams

2d

d) Ethoxyethane does not fully dissolve in water because:

- Hydrogen bonds exist between H_2O molecules; [1 mark]
- Hydrogen bonds cannot form between $C_2H_5OC_2H_5$ molecules; [1 mark]

[Total: 2 marks]

- The exam word 'suggest' means you might not have studied this specifically but need to apply your understanding
- To dissolve in a substance, the hydrogen bonds between water molecules need to break, and new hydrogen bonds form with the substance being dissolved
- If a substance does not fully dissolve, it is due to not enough hydrogen bonds between water and the substance forming
- Longer organic molecules are less soluble than smaller ones as the non-polar parts, in this case the C_2H_5 groups disrupt the hydrogen bonding

3a

a) Electronegativity is defined as:

- The ability of an atom to attract electrons / a pair of electrons to itself; [1 mark]
- In a covalent bond; [1 mark]

[Total: 2 marks]

- This is a standard definition that you are required to learn
- Sometimes the question may just be worth one mark, in which case both points would need to be included
- Although in a covalent bonds the electrons are shared between two atoms they are not always evenly spread
  - Some atoms are better at attracting electrons towards themselves than others in a covalent bond
  - The Pauling scale can be used to look at the electronegativity of an element
    - The more electronegative the element is, the higher the number and the better it is at attracting electrons to itself

3b

b) Explaining the trend in electronegativity across the Periodic Table:

- Electronegativity increases (across the Periodic Table; [1 mark]
- (Because) the nuclear charge increases
  AND
  Atomic radius decreases; [1 mark]
- (But) there is similar shielding; [1 mark]

[Total: 3 marks]

- The nuclear charge refers to the charge in the nucleus of the atom
- Going across a period / row in the Periodic Table, each element has one more proton than the one before it increasing the charge in the nucleus but the number of energy levels remains the same
- If the nuclear charge and atomic radius decreases, but shielding by inner electrons remains the same across the Periodic Table, electrons will be more strongly attracted to the nucleus of the atom
- Notice the question doesn't identify the trend so this should be your first marking point

3c

c) Explain how the carbon-hydrogen bond (such as in CH_4) differs from the nitrogen-hydrogen bond (such as in NH_3) in terms of the bond polarity:

- The nitrogen-hydrogen (N-H) bond is more polar than the carbon-hydrogen (C-H) bond; [1 mark]
- As the difference in electronegativity between the nitrogen and hydrogen atom is larger / 0.9
  OR
  The difference in electronegativity between the carbon and hydrogen atom is smaller / 0.4; [1 mark]

[Total: 2 marks]

- When there is an unequal distribution of electrons in a bond, we say that the bond is polar
- A bond is **more polar** when the difference between the electronegativities of the elements in the bond is **greater**
- Use the information in part a) to help you answer this question
- Don't just quote the numbers for nitrogen, hydrogen and carbon as this will not score the marks

3d

d) The bonding in ammonia (NH_3) is covalent but the bonding in lithium chloride (LiCl) is ionic because:

- The difference in electronegativity between nitrogen and hydrogen is small; [1 mark]
- So the electrons are shared; [1 mark]
- The difference in electronegativity between the lithium and chlorine atoms is large; [1 mark]
- So the electron is transferred from the lithium to the chlorine (to form ions); [1 mark]

[Total: 4 marks]

- As well as electronegativity, you need to demonstrate your knowledge of bonding in terms of electrons
  - Make an observation about the difference in electronegativity values of the two elements
  - Link this to what will happen to the electrons in the bond

- As the question has asked for your answer to be in terms of electronegativity, no credit will be awarded for explaining whether they are covalent or ionic based on whether they are metals or non-metals
- You must give the correct direction of electron transfer to score marking point 4

4a

a) The ions are held together in lithium and lithium chloride by:

- (Electrostatic) attraction / electrostatic forces between positive metal / lithium ions / Li^+ and (the sea of) delocalised electrons ; [1 mark]
- Electrostatic attraction / electrostatic forces between oppositely charged ions / Li^+ and Cl^-
  OR
  Electrostatic attraction / electrostatic forces between positive & negative ions; [1 mark]

[Total: 2 marks]

- This should be a straight forward question to answer as it covered at GCSE
- Correctly drawn diagrams would also score the marks
- The idea of attraction between ions with opposite charges must be clear

- Electrostatic attraction / electrostatic forces between oppositely charged ions / Li^+ and Cl^-
  OR
  Electrostatic attraction / electrostatic forces between positive & negative ions; [1 mark]

[Total: 2 marks]

- This should be a straight forward question to answer as it covered at GCSE
- Correctly drawn diagrams would also score the marks
- The idea of attraction between ions with opposite charges must be clear

4b

b) Explaining what can be deduced from the information in Table 4.1:

- The melting and boiling points of lithium chloride are higher than lithium
  OR
  The melting and boiling points of lithium are lower than lithium chloride
  OR
  The melting and boiling points of an ionic substance are higher than a metal

  OR

  The melting and boiling points of a metal are lower than an ionic substance; [1 mark]
- (Because) Ionic bonds (in lithium chloride) are stronger than metallic bonds (in lithium)
  OR
  Ionic bonds (or bonding) require more energy to break than metallic bonds
  OR
  Electrostatic attraction between oppositely charged ions (in lithium chloride) is stronger than electrostatic attraction between positive ions and delocalised electrons (in lithium metal); [1 mark]

[Total: 2 marks]

- Ionic bonds are stronger than metallic bonds and you are expected to know this for the exam
- In this question you have been asked to use the table, so you **must** use the table in your answer

- The more marks a question is worth, the more you will have to use the data and explain your answer in detail

4c

c) State and explain whether Student A, Student B, or neither student is correct:

- Both Student **A** and Student **B** are incorrect (neither student is correct)
  OR
  Student B is incorrect and Student A is partly correct; [1 mark]
- Lithium will conduct electricity (at all times); [1 mark]
- Lithium has delocalised electrons which can flow through the (metal) structure and carry the charge / carrying the current; [1 mark]
- Lithium chloride will conduct electricity when molten / dissolved in solution, but will not conduct electricity when solid; [1 mark]
- When solid the (negative) ions are fixed in position and cannot move
  AND
  When molten / dissolved in solution the (negative) ions are mobile and can move, carrying the charge / carrying current ; [1 mark]

[Total: 5 marks]

- For a substance to conduct electricity, there must be a flow of negative charge
  - I.e., a flow of electrons or a flow of negative ions

- A metal has delocalised electrons which can flow at all times; this is why metals conduct electricity
- Ionic solids will only conduct electricity when **molten/dissolved** in solution
  - When solid, the ions are fixed in their position and cannot move, so the negative ions cannot flow
  - No movement of negative ions, means no electrical conductivity
  - When molten/dissolved in solution the ions have been 'pulled apart' from their solid structure, and they are mobile (they can freely move around)
  - Because the ions are now able to flow, electricity can also flow

4d

d) Explain, in terms of electrons, how CaF_2 is formed from its atoms:

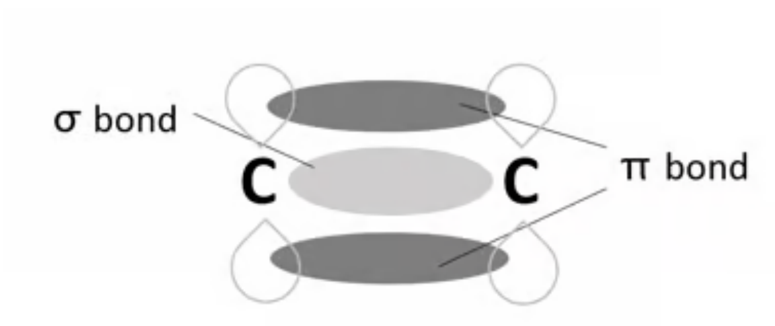
- One calcium atom loses two electrons; [1 mark]
- Two fluorine atoms each gain one electron; [1 mark]
- Forming one calcium ion / one Ca^{2+} and two fluoride ions / two F^-; [1 mark]

[Total: 3 marks]

- This question is worth 3 marks, so it is reasonable to deduce that you will score one mark for describing the loss / gain of electrons for each atom and the third for describing the ions produced
- Calcium is in Group 2 so will lose two electrons to gain a full outer shell and form a positively charged ion
- The formula tells you that there are two fluoride ions present, each of which will gain one electron to form a negatively charged ion

5a

a) The diagram to show the areas of electron density for each bond is:



- σ bond; [1 mark]
- π bond; [1 mark]

[Total: 2 marks]

- The π bond must be above and below the carbons but only one of the lobes of the π bond needs to be labelled
- **Remember:** A sigma bond is formed from the end to end overlap of s and p orbitals and pi bonds are formed from the sideways overlap of adjacent p orbitals
- The two electron clouds shown above represent **one** π bond
- if you draw both bonds correctly but label them the wrong way around you will score 1 mark

5b

b) The covalent bond between carbon atoms in ethene is stronger than in ethane because:

- The C=C bond is shorter (than the C-C bond); [1 mark]
- (Because) there is a larger electron density / more electrons between carbon atoms (in ethene); [1 mark]
- There is more attraction between the electrons and the atomic nuclei; [1 mark]

[Total: 3 marks]

- You must ensure you use the information from the table and make an observation about bond length
- **Remember:** The shorter the bond length, the stronger the covalent bond due to the increased forces of attraction between the electrons and atomic nucleus

5c

c) Comparing the reactivity of hydrogen halides:

- H-I is the most reactive and H-Cl is the least reactive
  OR
  The order of reactivity starting with the most reactive is HI > HBr > HCl; [1 mark]
- (Because) HI has the largest bond length and lowest bond energy; [1 mark]